

The straight line

Rule-1. The general equations of a straight line
 $a_1x + b_1y + c_1z + d_1 = a_2x + b_2y + c_2z + d_2 = 0$.

Rule-2. The equations of the symmetrical form of the straight line

$$\frac{x-x_1}{l} = \frac{y-y_1}{m} = \frac{z-z_1}{n}$$

Rule-3. The equations of a straight line which passes through a point (x_1, y_1, z_1) and whose direction ratios are a, b, c .

$$\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$$

Rule-4. Symmetrical form in terms of direction Cosines

Equations of x -axis $\frac{x-x_1}{1} = \frac{y-0}{0} = \frac{z-0}{0}$

Equations of y -axis $\frac{x-0}{0} = \frac{y-y_1}{1} = \frac{z-0}{0}$

Equations of z -axis $\frac{x-0}{0} = \frac{y-0}{0} = \frac{z-z_1}{1}$

Rule-5. Symmetrical form in terms of direction ratios

Equations of x -axis $\frac{x-x_1}{a} = \frac{y-0}{0} = \frac{z-0}{0}$

Equations of y -axis $\frac{x-0}{0} = \frac{y-y_1}{b} = \frac{z-0}{0}$

Equations of z -axis $\frac{x-0}{0} = \frac{y-0}{0} = \frac{z-z_1}{c}$

Rule-6. Standard equations of the three axes,

Equations of x-axis, $y=0=z$

Equations of y-axis, $x=0=z$

Equations of z-axis, $x=0=y$

Rule-7. Equations of a straight line through two points

(x_1, y_1, z_1) and (x_2, y_2, z_2) are

$$\frac{x-x_1}{x_1-x_2} = \frac{y-y_1}{y_1-y_2} = \frac{z-z_1}{z_1-z_2}$$

Rule-8. Direction ratios of a straight line

Let the equations of the straight line be

$$(a_1x + b_1y + c_1z + d_1 = 0 = a_2x + b_2y + c_2z + d_2)$$

the direction ratios of the line are

$$\begin{array}{ccccccc} a_1 & b_1 & -c_1 & -a_1 & -b_1 \\ & \swarrow & \searrow & \swarrow & \searrow \\ a_2 & b_2 & -c_2 & -a_2 & -b_2 \end{array}$$

$$b_1c_2 - b_2c_1, c_1a_2 - c_2a_1, a_1b_2 - a_2b_1$$

$$\therefore \text{The straight line } \frac{x}{b_1c_2 - b_2c_1} = \frac{y}{c_1a_2 - c_2a_1} = \frac{z}{a_1b_2 - a_2b_1}$$

Rule-9. The equations of the straight line are

$$a_1x + b_1y + c_1z + d_1 = 0 = a_2x + b_2y + c_2z + d_2$$

The direction ratios of the line

$$a_1 b_2 - b_1 a_2, a_2 c_1 - c_2 a_1, a_1 c_2 - c_1 a_2$$

$$b_1 c_2 - b_2 c_1, c_1 a_2 - c_2 a_1, a_1 b_2 - a_2 b_1$$

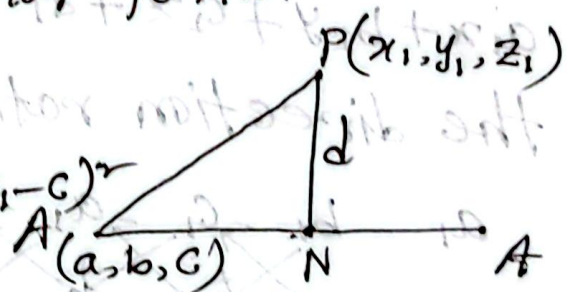
Rule-10. Distance of a point from a line

Let d be the length of the perpendicular distance from a point (x_1, y_1, z_1) to the line AB whose equations are $\frac{x-a}{l} = \frac{y-b}{m} = \frac{z-c}{n}$ with direction cosines l, m, n and passes through the point $A(a, b, c)$.

PN from P perpendicular to AB.

$$\therefore PN^2 = AP^2 - AN^2$$

$$AP^2 = (x_1 - a)^2 + (y_1 - b)^2 + (z_1 - c)^2$$



and $AN = \text{projection of AP on the given line AB}$
 $= l(x_1 - a) + m(y_1 - b) + n(z_1 - c)$

$$\therefore PN^2 = \{(x_1 - a)^2 + (y_1 - b)^2 + (z_1 - c)^2\}$$

$$- \{l(x_1 - a) + m(y_1 - b) + n(z_1 - c)\}^2$$

$$= d^2 = \left| \begin{array}{cc} x_1 - a & y_1 - b \\ 1 & m \end{array} \right|^2 + \left| \begin{array}{cc} y_1 - b & z_1 - c \\ m & n \end{array} \right|^2 + \left| \begin{array}{cc} z_1 - c & x_1 - a \\ n & l \end{array} \right|^2$$

$$\text{Let } \frac{x-a}{l} = \frac{y-b}{m} = \frac{z-c}{n} = r$$

$$\therefore N(lr+a, mr+b, nr+c)$$

Rule-11. Angle between two straight lines

Let the equations of symmetric straight lines be

$$\frac{x-x_1}{a_1} = \frac{y-y_1}{b_1} = \frac{z-z_1}{c_1} \longrightarrow (1)$$

$$\text{and } \frac{x-x_2}{a_2} = \frac{y-y_2}{b_2} = \frac{z-z_2}{c_2} \longrightarrow (2)$$

The direction ratios of the lines (1) and (2) are a_1, b_1, c_1 and a_2, b_2, c_2 respectively.

Let θ be the angle between the two lines (1) and

$$(2), \text{ then } \cos \theta = \frac{a_1 a_2 + b_1 b_2 + c_1 c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2} \cdot \sqrt{a_2^2 + b_2^2 + c_2^2}}$$

Condition of perpendicular: $a_1 a_2 + b_1 b_2 + c_1 c_2 = 0$

$$\text{Condition of parallel: } \frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

Rule-12. Angle between the plane and straight line

The angle between the plane $ax+by+cz+d=0$

$$\text{and the line } \frac{x-x_1}{a_1} = \frac{y-y_1}{b_1} = \frac{z-z_1}{c_1}$$

$$\cos \theta = \frac{\sqrt{(ab_1 - a_1b)^2 + (bc_1 - b_1c)^2 + (ca_1 - c_1a)^2}}{\sqrt{a^2 + b^2 + c^2} \cdot \sqrt{a_1^2 + b_1^2 + c_1^2}}$$

condition of parallel: $aa_1 + bb_1 + cc_1 = 0$

condition of perpendicular: $\frac{a_1}{a} = \frac{b_1}{b} = \frac{c_1}{c}$

Rule-13. Let the equations of the given lines be

$$\frac{x-x_1}{a_1} = \frac{y-y_1}{b_1} = \frac{z-z_1}{c_1}$$

$$\text{and } \frac{x-x_2}{a_2} = \frac{y-y_2}{b_2} = \frac{z-z_2}{c_2}$$

The condition of coplanar: $\begin{vmatrix} x_2-x_1 & y_2-y_1 & z_2-z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix} = 0$.

The equation of plane,

$$\begin{vmatrix} x-x_1 & y-y_1 & z-z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix} = 0 \quad \text{or} \quad \begin{vmatrix} x-x_2 & y-y_2 & z-z_2 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix} = 0$$

Rule-14. one line is symmetric form and the other in general form

Let the equations of the given lines be

$$\frac{x-x_1}{l_1} = \frac{y-y_1}{m_1} = \frac{z-z_1}{n_1}$$

$$\text{and } a_1x + b_1y + c_1z + d_1 = 0 = a_2x + b_2y + c_2z + d_2$$

The equation plane: $\frac{a_1x + b_1y + c_1z + d_1}{a_1l_1 + b_1m_1 + c_1n_1} = \frac{a_2x + b_2y + c_2z + d_2}{a_2l_1 + b_2m_2 + c_2n_2}$

Rule-15. The equations of a straight line intersecting two given lines

$$\frac{x-x_1}{a_1} = \frac{y-y_1}{b_1} = \frac{z-z_1}{c_1} = r_1 \rightarrow (1)$$

$$\text{and } \frac{x-x_2}{a_2} = \frac{y-y_2}{b_2} = \frac{z-z_2}{c_2} = r_2 \rightarrow (2)$$

From (1) $\Rightarrow (r_1a_1 + x_1, b_1r_1 + y_1, c_1r_1 + z_1)$

From (2) $\Rightarrow (a_2r_2 + x_2, b_2r_2 + y_2, c_2r_2 + z_2)$

Rule-16. The equations of the given lines be

$$\frac{x-x_1}{l_1} = \frac{y-y_1}{m_1} = \frac{z-z_1}{n_1}$$

$$\text{and } \frac{x-x_2}{l_2} = \frac{y-y_2}{m_2} = \frac{z-z_2}{n_2}$$

the line of shortest distance between the two

$$\text{lines, } \left| \begin{array}{ccc} x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \end{array} \right| \div \sqrt{\sum (m_1n_2 - m_2n_1)^2}$$

Rule-17. The equations of the line of shortest distance are

the equation plane: $\begin{vmatrix} x-x_1 & y-y_1 & z-z_1 \\ 1 & m_1 & n_1 \end{vmatrix} = 0$

the given lines $\begin{vmatrix} x-x_2 & y-y_2 & z-z_2 \\ 1 & m_2 & n_2 \end{vmatrix} = 0$

and $\frac{x-x_1}{a_1} = \frac{y-y_1}{b_1} = \frac{z-z_1}{c_1}$

From (1) $\Rightarrow (x_1 + a_1, y_1 + b_1, z_1 + c_1)$

From (2) $\Rightarrow (x_2 + a_2, y_2 + b_2, z_2 + c_2)$

the equations of the given lines are

$\frac{x-x_1}{a_1} = \frac{y-y_1}{b_1} = \frac{z-z_1}{c_1}$

and $\frac{x-x_2}{a_2} = \frac{y-y_2}{b_2} = \frac{z-z_2}{c_2}$

the line of shortest distance between the two

line? $\begin{vmatrix} x-x_1 & y-y_1 & z-z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix} \div \sqrt{a_1^2 + b_1^2 + c_1^2}$

the equations of the line of shortest

distance are

problem-1: Find the symmetrical form the equation of the line $x-4y+2z+7=0=3x+3y-z-2$.

solution: The given equations of the line are

$$x-4y+2z+7=0 \longrightarrow (1)$$

$$3x+3y-z-2=0 \longrightarrow (2)$$

Let a, b, c be the direction ratios of the line
(1) & (2) $\Rightarrow$

$$\begin{array}{ccc} 1 & -4 & 2 \\ 3 & 3 & -1 \end{array} \quad \begin{array}{ccc} 2 & -1 & 3 \\ 1 & 3 & -4 \end{array} \quad \begin{array}{ccc} 1 & -4 & 2 \\ 3 & 3 & -1 \end{array}$$

$$\therefore a = -2, b = 7, c = 15$$

Now to find the co-ordinates of a point on the line given by (1) & (2), let us find the point where the line meets the xy -plane provided the line is not parallel to the xy -plane.

Putting $z=0$ in the equations given by (1), we have, $x-4y+7=0 \longrightarrow (3)$

$$3x+3y-2=0 \longrightarrow (4)$$

Solving these, we get $x = -\frac{13}{15}, y = \frac{23}{15}$

The line meets the plane $z=0$ in the point

$$\left(-\frac{13}{15}, \frac{23}{15}, 0\right).$$

Therefore, the equations of the given line in symmetric form are

$$\frac{x + \frac{13}{15}}{-2} = \frac{y - \frac{23}{15}}{7} = \frac{z - 0}{15} \quad \underline{\underline{Ans}}$$

Alternative method: If yz -plane, $x=0$.

$$\text{From (1) \& (2)} \Rightarrow -4y + 2z + 7 = 0 \quad \text{--- (c) \& (d)}$$

$$3y - z - 2 = 0$$

$$\text{Solving these, we get } y = -\frac{3}{2}, z = -\frac{13}{2}$$

$$\therefore \text{Symmetric form, } \frac{x-0}{-2} = \frac{y + \frac{3}{2}}{7} = \frac{z + \frac{13}{2}}{15} \quad \underline{\underline{Ans}}$$

Alternative method: If zx -plane, $y=0$.

$$\text{From (1) \& (2)} \Rightarrow x + 2z + 7 = 0$$

$$3x - z - 2 = 0$$

$$\text{Solving these, we get } x = -\frac{3}{7}, z = -\frac{23}{7}$$

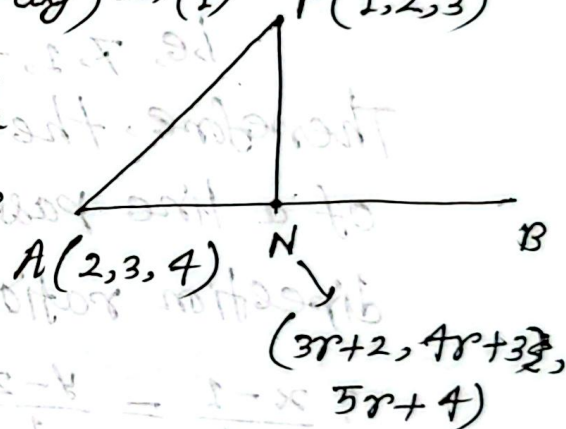
$$\therefore \text{Symmetric form, } \frac{x + \frac{3}{7}}{-2} = \frac{y-0}{7} = \frac{z + \frac{23}{7}}{15}$$

problem-2: From the point $P(1, 2, 3)$, PN is drawn perpendicular to the straight line $\frac{x-2}{3} = \frac{y-3}{4} = \frac{z-4}{5}$. Find the distance PN , the equations to PN and the co-ordinates of N .

Solution: The equations of the given line AB are

$$\frac{x-2}{3} = \frac{y-3}{4} = \frac{z-4}{5} = r \text{ (say)} \rightarrow (1)$$

The line (1) is passing through the point $A(2, 3, 4)$. Since N , the foot of the perpendicular,



is a point on the line AB ,

the co-ordinates of N may be written as

$$(3r+2, 4r+3, 5r+4) \rightarrow (2)$$

The direction ratios of PN are

$$3r+2-1, 4r+3-2, 5r+4-3$$

i.e., $3r+1, 4r+1, 5r+1 \rightarrow (3)$ and DR's of AB are

since PN is perpendicular to AB , so we have

$$3 \cdot (3r+1) + 4 \cdot (4r+1) + 5 \cdot (5r+1) = 0$$

$$\Rightarrow r = -\frac{6}{25}$$

putting the values of r in (2), we get

$$N\left(\frac{32}{25}, \frac{51}{25}, \frac{14}{5}\right)$$

$$\begin{aligned}\therefore PN &= \sqrt{\left(\frac{32}{25} - 1\right)^2 + \left(\frac{51}{25} - 2\right)^2 + \left(\frac{14}{5} - 3\right)^2} \\ &= \sqrt{\frac{49}{625} + \frac{1}{625} + \frac{1}{25}} \\ &= \frac{\sqrt{3}}{5}\end{aligned}$$

putting the value of r in (3), the DR's of PN are $\frac{7}{25}, \frac{1}{25}, -\frac{5}{25}$

i.e. 7, 1, -5

Therefore, the equations to PN are, the equ. of a line passing through $P(1, 2, 3)$ and having direction ratios 7, 1, -5.

$$\therefore \frac{x-1}{7} = \frac{y-2}{1} = \frac{z-3}{-5}$$

Problem-3: Show that the line $\frac{x+3}{2} = \frac{y+5}{3} = \frac{z-7}{-3}$

and the line $\frac{x+1}{4} = \frac{y+1}{5} = \frac{z+1}{-1}$ are coplanar.

Also find the equation of the plane containing them.

Solution: We know that the two lines

$$\frac{x-x_1}{a_1} = \frac{y-y_1}{b_1} = \frac{z-z_1}{c_1} \text{ and } \frac{x-x_2}{a_2} = \frac{y-y_2}{b_2} = \frac{z-z_2}{c_2}$$

Will be coplanar if
$$\begin{vmatrix} x_2-x_1 & y_2-y_1 & z_2-z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix} = 0$$

The given two lines are

$$\frac{x+3}{2} = \frac{y+5}{3} = \frac{z-7}{-3} \longrightarrow (1)$$

$$\text{and } \frac{x+1}{4} = \frac{y+1}{5} = \frac{z+1}{-1} \longrightarrow (2)$$

Here, $x_1 = -3, y_1 = -5, z_1 = 7, x_2 = -1, y_2 = -1, z_2 = -1$

$$a_1 = 2, b_1 = 3, c_1 = -3, a_2 = 4, b_2 = 5, c_2 = -1.$$

$$\text{Now, } \begin{vmatrix} -1+3 & -1+5 & -1-7 \\ 2 & 3 & -3 \\ 4 & 5 & -1 \end{vmatrix}$$

$$= \begin{vmatrix} 2 & 4 & -8 \\ 2 & 3 & -3 \\ 4 & 5 & -1 \end{vmatrix} = 2(-3+15) - 4(-2+12) - 8(10-12)$$

$$= 24 - 40 + 16 = 0.$$

Hence the given two lines are coplanar.

The equation of the plane containing the lines (1)

$$\text{and (2) is } \begin{vmatrix} x+3 & y+5 & z-7 \\ 2 & 3 & -3 \\ 4 & 5 & -1 \end{vmatrix} = 0$$

$$\Rightarrow (x+3) \cdot 12 - (y+5) \cdot 10 + (z-7) \cdot (-2) = 0$$

$$\Rightarrow 6x - 5y - z = 0 \quad \underline{\text{Ans.}}$$

Problem-4: prove that the lines $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and $\frac{x-2}{3} = \frac{y-3}{4} = \frac{z-4}{5}$ are coplanar; find their point of intersection. Also find the equation of the plane in which they lie.

Solution: The equations of the given lines are $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4} = r_1$ (say) $\rightarrow (1)$

and $\frac{x-2}{3} = \frac{y-3}{4} = \frac{z-4}{5} = r_2$ (say) $\rightarrow (2)$

The co-ordinates of any point P on the line (1) are $(2r_1+1, 3r_1+2, 4r_1+3)$ and those of a point Q on the line (2) are $(3r_2+2, 4r_2+3, 5r_2+4)$. If the line (1) and (2) are coplanar then they intersect, then for some values of r_1 and r_2 , the point P and Q should coincide, then we have

$$2r_1+1 = 3r_2+2 \Rightarrow 2r_1 - 3r_2 - 1 = 0 \rightarrow (3)$$

$$3r_1+2 = 4r_2+3 \Rightarrow 3r_1 - 4r_2 - 1 = 0 \rightarrow (4)$$

$$4r_1+3 = 5r_2+4 \Rightarrow 4r_1 - 5r_2 - 1 = 0 \rightarrow (5)$$

From (3) & (4), we get $r_1 = -1, r_2 = -1$.

These values of r_1 and r_2 satisfy the equation (5), since $4 \cdot (-1) - 5 \cdot (-1) - 1 = -4 + 5 - 1 = 0$.

substituting the value of r_1 or r_2 the co-ordinates of the point of intersection P or Q are $(-1, -1, -1)$ or $(-1, -1, -1)$.

The equation of the plane containing the lines (1) and (2) is $\begin{vmatrix} x-1 & y-2 & z-3 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{vmatrix} = 0$

$$\Rightarrow x - 2y + z = 0. \quad \underline{\underline{\text{Ans}}}$$

Problem-5: Show that the line $\frac{x+4}{3} = \frac{y+6}{5} = \frac{z-1}{-2}$ and the line $3x - 2y + z + 5 = 0 = 2x + 3y + 4z - 4$ are coplanar. Also find the equation of the plane in which they lie and find the point of intersection of the two lines.

Solution: The equations of the given lines are

$$\frac{x+4}{3} = \frac{y+6}{5} = \frac{z-1}{-2} = r \text{ (say)} \longrightarrow (1)$$

$$\text{and } 3x - 2y + z + 5 = 0 = 2x + 3y + 4z - 4 \longrightarrow (2)$$

The equation of any plane through the line (2) is

$$(3x - 2y + z + 5) + k(2x + 3y + 4z - 4) = 0 \longrightarrow (3)$$

$$\Rightarrow (2k+3)x + (3k-2)y + (4k+1)z + (5-4k) = 0 \longrightarrow (4)$$

If the plane (4) will be perpendicular to the line (1) we get,

$$3 \cdot (2k+3) + 5 \cdot (3k-2) - 2(4k+1) = 0$$

$$\Rightarrow k = \frac{3}{13}$$

Putting the values of k in (3) $\Rightarrow$

$$(3x-2y+z+5) + \frac{3}{13} (2x+3y+4z-4) = 0$$

$$\Rightarrow 45x - 17y + 25z + 53 = 0 \longrightarrow (5)$$

The line (1) passes through the point $(-4, -6, 1)$ and it satisfies the equation (5) as

$$45 \cdot (-4) - 17 \cdot (-6) + 25 \cdot (1) + 53$$

$$= -180 + 102 + 25 + 53$$

$$= 0.$$

Hence the lines (1) and (2) are coplanar and they lie in the plane $45x - 17y + 25z + 53 = 0$

To find the point of intersection,

$$\text{From (1)} \Rightarrow (3r-4, 5r-6, -2r+1)$$

The lines (1) and (2) will be coplanar, if this point also lie in the line (2), i.e. if the point satisfies both the equations of the line (2).

Hence we have,

$$3 \cdot (3r-4) - 2 \cdot (5r-6) + 1 \cdot (-2r+1) + 5 = 0$$

$$\Rightarrow r = 2$$

$\therefore$ The point of intersection is $(2, 4, -3)$.

Problem-6: Find the shortest distance between the lines $\frac{x-1}{4} = \frac{y-2}{3} = \frac{z-1}{-5}$, $\frac{x+1}{2} = \frac{y-3}{3} = \frac{z-4}{-4}$ and the equation containing them.

Solution: The given lines are

$$\frac{x-1}{4} = \frac{y-2}{3} = \frac{z-1}{-5} \rightarrow (1)$$

$$\frac{x+1}{2} = \frac{y-3}{3} = \frac{z-4}{-4} \rightarrow (2)$$

Let, l, m, n be the direction cosines of the shortest distance line. Since the shortest distance line is perpendicular to both the lines (1) and (2), so we have

$$4l + 3m - 5n = 0$$

$$2l + 3m - 4n = 0$$

$$\therefore \frac{l}{3} = \frac{m}{6} = \frac{n}{6}$$

$$\Rightarrow \frac{l}{1} = \frac{m}{2} = \frac{n}{2} = \frac{\sqrt{1^2 + 2^2 + 2^2}}{\sqrt{1^2 + 2^2 + 2^2}} = \frac{1}{3}$$

$$\therefore l = \frac{1}{3}, m = \frac{2}{3}, n = \frac{2}{3}$$

The lines (1) and (2) pass through the points $(1, 2, 1)$ and $(-1, 3, 4)$ respectively.

S.D. = Projection of the line joining the two points $(1, 2, 1)$ and $(-1, 3, 4)$ on the S.D. line

$$= \frac{1}{3}(-1-1) + \frac{2}{3}(3-2) + \frac{2}{3}(4-1)$$

$$= -\frac{2}{3} + \frac{2}{3} + \frac{2}{3} \cdot 3$$

$$= 2 \quad \underline{\underline{Ans}}$$

The equation of the plane which contains the line (1) and the S.D. line is

$$\begin{vmatrix} x-1 & y-2 & z-1 \\ 4 & 3 & -5 \\ 1 & 2 & 2 \end{vmatrix} = 0$$

$$\Rightarrow (x-1) \cdot 16 - (y-2) \cdot 13 + (z-1) \cdot 5 = 0$$

$$\Rightarrow 16x - 13y + 5z + 5 = 0 \quad \underline{\underline{Ans}}$$

The equation of the plane which contains the line (2) and the S.D. line is

$$\begin{vmatrix} x+1 & y-3 & z-4 \\ 2 & 3 & -4 \\ 1 & 2 & 2 \end{vmatrix} = 0$$

$$\Rightarrow (x+1) \cdot 14 - (y-3) \cdot 8 + (z-4) \cdot 1 = 0$$

$$\Rightarrow 14x - 8y + z + 34 = 0 \quad \underline{\underline{Ans}}$$

problem-7: Find the point of intersection of the line $\frac{x-2}{3} = \frac{y+1}{4} = \frac{z-2}{12}$ and the plane $x-2y+z-13=0$.

solution: The given equations are

$$\frac{x-2}{3} = \frac{y+1}{4} = \frac{z-2}{12} \longrightarrow (1)$$

$$x-2y+z-13=0 \longrightarrow (2)$$

From (1), we get

$$\frac{x-2}{3} = \frac{y+1}{4} = \frac{z-2}{12} = r \text{ (say)}$$

The co-ordinates of any point P on the line (1) are $(3r+2, 4r-1, 12r+2) \longrightarrow (3)$

suppose the line (1) intersect the plane (2) in this point P, so the point P lies on the plane (2), then we have

$$(3r+2) - 2(4r-1) + (12r+2) - 13 = 0$$

$$\Rightarrow r=1$$

putting the value of r in the co-ordinates of P, the required co-ordinates of the point of intersection are given by

$$(3 \cdot 1 + 2, 4 \cdot 1 - 1, 12 \cdot 1 + 2)$$

$$\text{i.e., } (5, 3, 14) \underline{\underline{A \cdot \cdot}}$$

Problem-8: Find the distance of the point $(1, 1, 1)$ from the point of intersection of the line $\frac{x-1}{2} = \frac{y-1}{3} = \frac{z-1}{6}$ and the plane $4x - 3y + 2z + 6 = 0$.

Solution: The given equations are

$$\frac{x-1}{2} = \frac{y-1}{3} = \frac{z-1}{6} = r \text{ (say)} \rightarrow (1)$$

$$4x - 3y + 2z + 6 = 0 \rightarrow (2)$$

From (1), we have

$$(2r+1, 3r+1, 6r+1)$$

Now, (1) intersect the plane (2) $\Rightarrow$

$$4 \cdot (2r+1) - 3 \cdot (3r+1) + 2 \cdot (6r+1) + 6 = 0$$

$$\Rightarrow r = -\frac{2}{11}$$

$\therefore$ The point of intersection are $(-\frac{7}{11}, -\frac{16}{11}, -\frac{43}{11})$

Therefore, the distance between the points

$(1, 1, 1)$ and $(-\frac{7}{11}, -\frac{16}{11}, -\frac{43}{11})$ is

$$= \sqrt{\left(-\frac{7}{11} - 1\right)^2 + \left(-\frac{16}{11} - 1\right)^2 + \left(-\frac{43}{11} - 1\right)^2}$$

$$= \frac{63}{11} \text{ Ans.}$$

Problem-9: Show that the lines $\frac{x-5}{4} = \frac{y-7}{4} = \frac{z+3}{-5}$ and $\frac{x-8}{7} = \frac{y-4}{1} = \frac{z-5}{3}$ are coplanar and the equation of the plane containing them is $17x - 47y - 24z + 172 = 0$. $\rightarrow$ For practice.

Problem-10: Show the straight lines $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and $\frac{x-2}{3} = \frac{y-3}{4} = \frac{z-4}{5}$ are coplanar and the equation of the plane containing them is $x - 2y + z = 0$. $\rightarrow$ For practice.

Problem-11: Find the distance of the point $(1, 1, 1)$ from the point where the line

$$\frac{x-1}{2} = \frac{y-1}{3} = \frac{z-2}{6} \text{ cuts plane } 4x - 3y + 2z + 6 = 0.$$

Problem-12: Find the length and equation of shortest distance between two straight lines

$$\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-4}{4} \text{ and } \frac{x-2}{3} = \frac{y-3}{4} = \frac{z-4}{5}.$$

Problem-13: Find the length and equation of shortest distance between two straight lines

$$\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-4}{4} \text{ and } \frac{x-2}{2} = \frac{y-4}{4} = \frac{z-5}{5}.$$